



HOW DOES A RANDOM FOREST WORK?

Now that you have a dataset of cleaned and engineered features, it is time to build a predictive model to see how well these features are able to predict a customer churning.

Estelle has informed you that a classification model would be best for this task and has suggested that you try the **Random Forest classifier**.

A random forest is a supervised learning algorithm which means that you must provide the algorithm with a set of features, as well as the outcome that you're trying to predict, in our case churn.

Random Forest classifiers make predictions by building a set of decision trees on different samples of the data and by taking a majority vote to decide what prediction to make.

To visualize this, the image below shows 9 decision trees and they are all trying to predict an outcome which is either a 1 or a 0 (similar to our case, where if someone has churned you see a 1, and if they haven't you see a 0).

The random forest would look at all the predictions generated from the 9 trees. You can see that 6 trees have predicted 1 and 3 have predicted 0. Therefore, the random forest would take the majority vote and present its prediction as equal to 1.



OUTLINE FOR MAKING YOUR PREDICTIONS

After the previous task of feature engineering, Estelle conducted a further review and has provided you with a final dataset to use this for this task, named “data_for_predictions.csv”. Be sure to use this dataset for this task.

To best predict the customer churn, your task is to:

- Train a random forest classifier to predict churn
- Evaluate the predictions using metrics to demonstrate how accurately the model has performed

Estelle has provided a Jupyter notebook to get started. You should use this as a template to complete the work for this task.

You will notice that within the notebook, Estelle has imported various packages that would be used. One of them is named “scikit-learn”. This is an open source machine learning package and will be the source of the random forest model, as well as other things, that we use.

For more information on how to use the Random Forest classifier in scikit-learn, see the documentation site [here](#)

The outputs of your work will be shared with the AD and Estelle has given you a few points to include within the notebook:

- Why did you choose the evaluation metrics that you used? Please elaborate on your choices.
- Do you think that the model performance is satisfactory? Give justification for your answer.
- Make sure that your work is presented clearly with comments and explanations

EXAMPLE ANSWER

A few notes on model evaluation:

- It was left for you to decide how to evaluate the performance of the model. In general, you want to use metrics that reflect honestly how well the model has performed.
- In the example notebook we use 3 metrics, accuracy, precision and recall.
- The reason why we are using these three metrics is because a simple accuracy measure (what percentage was predicted correctly) is not always a good measure to use when there isn't an equal split between categories (e.g., churn is the category we care most about but is only 10% of the data).
- After calculating the 3 metrics, we can see that we're able to accurately identify clients that do not churn, but not so accurately identify clients that will churn. Our model is predicting a high percentage of clients to not churn, when in fact they did!
- This suggests that more work needs to be done on feature engineering and model tuning to reduce the number of false negatives in our predictions.

Finally, we produce a feature importance chart to visualize which features were indeed useful within the model and which ones weren't.

- We can see that net margin and consumption over 12 months were important predictors of churn.
- However the price sensitivity features are scattered around and do not shine through as a main driver for churn in their current form.